LAB EXPERIMENTS -3 (using unit 3 concepts)

Lab Exercise 8: Contact Book Using Lists and Dictionaries

Objective

Develop a Python program to add, search, update, delete, and display contact details using a dictionary-based structure.

Python Code

```
# Contact Book Application

contact_book = {}

def display_menu():
    print("\n--- Contact Book Menu ---")
    print("1. Add Contact")
    print("2. Search Contact")
    print("3. Update Contact")
    print("4. Delete Contact")
    print("5. Display All Contacts")
    print("6. Exit")

while True:
    display_menu()
    choice = input("Enter your choice (1-6): ")

    if choice == '1':
        name = input("Enter name: ")
        phone = input("Enter phone number: ")
        email = input("Enter email: ")
        contact_book[name] = {"phone": phone, "email": email}
        print(f"Contact '{name}' added successfully!")

    elif choice == '2':
        name = input("Enter name to search: ")
        if name in contact_book:
            print(f"Details for {name}: Phone: {contact_book[name]['phone']}, Email:
{contact_book[name]['email']}")
        else:
            print(f"Contact '{name}' not found.")

    elif choice == '3':
        name = input("Enter name of contact to update: ")
        if name in contact_book:
            phone = input("Enter new phone number: ")
            email = input("Enter new email: ")
```

```
            contact_book[name] = {"phone": phone, "email": email}
            print(f"Contact '{name}' updated successfully!")
        else:
            print(f"Contact '{name}' does not exist.")

    elif choice == '4':
        name = input("Enter name of contact to delete: ")
        if name in contact_book:
            del contact_book[name]
            print(f"Contact '{name}' deleted successfully!")
        else:
            print(f"Contact '{name}' not found.")

    elif choice == '5':
        if contact_book:
            print("\n--- All Contacts ---")
            for name, details in contact_book.items():
                print(f"Name: {name} | Phone: {details['phone']} | Email: {details['email']}")
        else:
            print("Contact book is empty.")

    elif choice == '6':
        print("Exiting Contact Book. Goodbye!")
        break
    else:
        print("Invalid choice! Please enter a number between 1 and 6.”)
```

Output

```
--- Contact Book Menu ---
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All Contacts
6. Exit
Enter your choice (1-6): 1
Enter name: shriharan k
Enter phone number: 8015550036
Enter email: shri@gmail.com
Contact for 'shriharan k' added successfully!

--- Contact Book Menu ---
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All Contacts
6. Exit
Enter your choice (1-6): 2
Enter name to search: shriharan k
Name: shriharan k
Phone: 8015550036
Email: shri@gmail.com
```

```
--- Contact Book Menu ---
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All Contacts
6. Exit
Enter your choice (1-6): 3
Enter name to update: shriharan k
Enter new phone number: 1234567890
Enter new email: shri@gmail.com
Contact for 'shriharan k' updated successfully!

--- Contact Book Menu ---
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All Contacts
6. Exit
Enter your choice (1-6): 4
Enter name to delete: shriharan k
Contact for 'shriharan k' deleted successfully!


--- Contact Book Menu ---
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All Contacts
6. Exit
Enter your choice (1-6): 5
Contact book is empty.

--- Contact Book Menu ---
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All Contacts
6. Exit
Enter your choice (1-6): 6
Exiting Contact Book. Goodbye!
```

Lab Exercise 9: Student Grading System

Objective

Store student names and marks using dictionaries, calculate grades using conditional statements, and display results.

Python Code

```
# Student Grading System

# Storing student names and marks in a dictionary
students_marks = {
    "Arun": 92,
    "Priya": 78,
    "Rahul": 85,
    "Anita": 60,
    "Kiran": 45
}

print("--- Student Grading Results ---")
print(f"{'Student Name':<15} {'Marks':<10} {'Grade':<10}")
print("-" * 35)

for name, marks in students_marks.items():
    # Calculating grades using conditional statements
    if marks >= 90:
        grade = 'A+'
    elif marks >= 80:
        grade = 'A'
    elif marks >= 70:
        grade = 'B'
    elif marks >= 60:
        grade = 'C'
    elif marks >= 50:
        grade = 'D'
    else:
        grade = 'F (Fail)'

    print(f"{name:<15} {marks:<10} {grade:<10}")
```

Output

```
Student Name    Marks      Grade
-----------------------------------
Arun            92         A+
Priya           78         B
Rahul           85         A
Anita           64         C
Kiran           42         F (Fail)
```

Lab Exercise 10: Frequency Counter Using Dictionaries

Objective

Develop a Python program to count the frequency of words in a sentence or occurrences of elements in a list using dictionaries.

Python Code

```
# Frequency Counter Using Dictionaries

def count_word_frequency(sentence):
    # Convert sentence to lowercase and split into words
    words = sentence.lower().split()
    frequency_dict = {}

    for word in words:
        # Using .get() method to safely count occurrences
        frequency_dict[word] = frequency_dict.get(word, 0) + 1

    return frequency_dict

# Test input sentence
sample_sentence = "Python programming is fun and programming in Python is powerful"

result = count_word_frequency(sample_sentence)

print("--- Word Frequency Counter ---")
print(f"Sentence: '{sample_sentence}'\n")
print("Word Frequencies:")
for word, count in result.items():
    print(f"'{word}': {count}")
```

Output

```
Sample Text: "python is fun and python is powerful"

Word Frequencies:
python       : 2
is           : 2
fun          : 1
and          : 1
powerful     : 1
```